How Shared Regularization Changes Calibration: A Case Study with Splines

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Abstract

Sharing spline coefficients changes regularization without expanding the function family. We show how the common penalty depends on squared class count and test the resulting predictions in multiclass calibration. Four prespecified fixed-penalty predictions are supported. In controlled synthetic experiments, independent tuning reduces the observed shared–class-only gap by 29% for common signals and 97.5% for heterogeneous signals. The latter still benefits despite zero mean class amplitudes, limiting a scalar explanation. Real-data comparisons do not establish general superiority. This case study connects shared regularization to its empirical benefits, tuning dependence, and limits.

1 The Question and the Findings

Post-hoc calibration learns a transformation of a frozen classifier's probabilities from separate labeled data. Temperature scaling (TS), matrix scaling, and splines offer different levels of flexibility [4, 7]. Our hierarchical shared-spline calibrator (HSSC) adds a common nonlinear correction and class-specific corrections to a structured matrix-scaling (SMS) baseline [1].

Class-specific splines already express a common correction. Sharing changes which responses the penalty suppresses, motivating a controlled regularization comparison rather than attribution from an overall gain [5, 6].

We test common, absent, and opposing class signals. HSSC is our own candidate, not a widely adopted method being overturned; shared/deviation regularization is established [3].

2 What Sharing Changes

A compact model. Let z_i be fixed log probabilities for example i, with K classes. Six fixed spline features ϕ_{ic} describe class c's gap from the highest score. In the unprojected setting studied here,

$$s_{ic} = \ell_c(z_i) + \theta^\top \phi_{ic} + U_c^\top \phi_{ic}, \qquad q_i = \mathrm{softmax}(s_i). \tag{1}$$

The affine correction $\ell_c(z_i)$ has a global slope, a full matrix, and a bias. The shared coefficients θ apply the same response to each class's gap; U_c allows departures. Mean negative log likelihood is minimized with

$$R = R_{\text{aff}} + \frac{J}{n}\left(\lambda_s \|\theta\|^2 + \lambda_c K \sum_c \|U_c\|^2\right), \quad J = 6. \tag{2}$$

Here n is calibration size. All penalties are positive; R_{aff} is the same positive quadratic affine penalty across controls. Fixed features give a convex NLL, and the positive penalty makes the complete objective strictly convex with a unique optimum. Full definitions and weights are in the supplied supplement, §2.

The central relationship. At any fixed affine correction, set $V_c = U_c + \theta$. This absorbs sharing into class-only coefficients; conversely, setting $\theta = 0$ realizes every V. Thus the function families coincide. Write $V_c = \mu + D_c$, where μ is the class mean and $\sum_c D_c = 0$. Compared with class-only ridge, sharing leaves the deviation penalty unchanged and multiplies the common penalty by

$$\boxed{s_K = \frac{\lambda_s}{\lambda_s + \lambda_c K^2}}. \tag{3}$$

Sharing makes the common response cheaper, with a class-count-dependent discount. At $\lambda_s = 10, \lambda_c = 1$, it retains 38.5% of the class-only common penalty for $K = 4$, but only 3.8% for $K = 16$. These percentages describe penalty weights, not predicted performance gains.

Short proof. Holding V fixed gives $\sum_c \|U_c\|^2 = \sum_c \|D_c\|^2 + K\|\mu - \theta\|^2$. Minimizing the resulting positive quadratic in θ gives $\theta_* = (1 - s_K)\mu$. Substitution leaves coefficient $\lambda_c KJ/n$ on $\sum_c \|D_c\|^2$ and $s_K \lambda_c K^2 J/n$ on $\|\mu\|^2$, proving the claim. Supplement §4 gives the projected variant with a free global slope.

From a penalty to a prediction. Weaker common shrinkage preserves useful shared corrections but admits more noise. In a scalar quadratic model with separable coordinates, it helps when reduced squared bias exceeds added variance; without common signal,

Table 1: **Do shrinkage predictions transfer to calibration risk?** New synthetic experiment; all differences are expected Brier times 10^4. Sharing uses $\lambda_s = 10$; scaled sharing uses $10(K/4)^2$. Negative differences favor the first method.

Regime	Comparison	Class-penalty choice	Mean difference	Interval
Prespecified predictions: 98.75% intervals, four-comparison Bonferroni adjustment				
Common signal	Shared – class-only	Fixed	-2.06	$[-2.63, -1.55]$
Null signal	Shared – class-only	Fixed	$+0.35$	$[0.03, 0.68]$
Null, $K = 8, 16$	Shared – scaled	Fixed	$+0.38$	$[0.06, 0.71]$
Common, $K = 8, 16$	Shared – scaled	Fixed	-1.88	$[-2.50, -1.33]$
Secondary comparisons: pointwise 95% intervals; five candidates per tuned method				
Common signal	Shared – class-only	Tuned	-1.46	$[-1.97, -0.98]$
Null signal	Shared – class-only	Tuned	$+0.30$	$[-0.09, 0.68]$
Class-specific signal	Shared – class-only	Fixed	-93.32	$[-95.20, -91.37]$
Class-specific signal	Shared – class-only	Tuned	-2.30	$[-2.49, -2.14]$

Paired bootstrap resamples 24 generator seeds (20,000 draws), averaging stated class counts and two truth shapes within seeds. Fixed means $\lambda_c = 1$. A null effect after tuning is unresolved, not proved absent. The heterogeneous regime has zero mean class response amplitudes; its gains limit a naive transfer of the scalar risk rule.

only the variance cost remains (derivation: supplement §14).

This motivates predictions, not a softmax risk theorem. We also test a scaling intervention: $\lambda_s = 10(K/4)^2$ holds the relative common penalty constant across K at each λ_c, retaining 38.5% at $\lambda_c = 1$. Stronger shrinkage can reduce noise fitting while suppressing useful signal.

3 An Experiment That Tests the Mechanism

Controlled probabilities and signals. Before generating outcomes, we locally fixed the generator, hypotheses, seeds, grids, and analysis; this was not external preregistration. For $K = 4, 8, 16$, log-normalize independent $N(0, 1.5^2)$ logits to z_i. With $r_{ic} = z_{ic} - \max_d z_{id}$, generate

$$p_i^* = \text{softmax}(s_i^*), \qquad s_{ic}^* = z_{ic} + (a + d_c) f(r_{ic}). \quad (4)$$

The response is either $f(r) = \text{clip}(r/2, -1, 0)$, which the fitted hinges can represent, or the smooth $\tanh(r/2)$, which they cannot represent exactly.

Null uses $a = d_c = 0$ (already correct probabilities); common uses $a = 4, d_c = 0$; heterogeneous uses $a = 0$ and balanced, randomly permuted $d_c = \pm 4$ (opposing responses). Each of 24 independent seeds supplies $32K$ labeled calibration examples and 8192 independent test inputs. The 432 records contain 360 distinct problems because null shapes coincide. Seeds are the statistical units.

Matched comparisons and tuning. The same affine and six-spline family is used without TS preprocessing or projection, isolating Equation (3). Class-only, shared $\lambda_s = 10$, and scaled sharing all use $\lambda_a = 1$. Compare fixed $\lambda_c = 1$ with independent three-fold Brier selection from $\{.01, .1, 1, 10, 100\}$. Methods use the same candidates and folds, then refit on all calibration examples. This finite search is not exhaustive.

Test *excess Brier* averages $\|q_i - p_i^*\|^2$ over held-out inputs. Irreducible outcome variance cancels in contrasts, giving the same differences as expected total Brier. Only sampled labels enter fitting and selection; known probabilities enter evaluation.

Predictions and uncertainty. We predicted that fixed-penalty sharing helps common signals and harms null signals, with scaling reducing both effects at $K = 8, 16$. Table 1 tests all four contrasts using 20,000 paired seed-block bootstrap draws and Bonferroni-adjusted 98.75% intervals, averaging specified class counts and shapes within seeds. Other contrasts use secondary, pointwise 95% intervals.

4 What the Results Mean

A common signal makes sharing useful. All four adjusted intervals support the predicted signs. Common-signal sharing reduces Brier by 2.06×10^{-4}, or 2.59% of excess Brier, consistent with less shrinkage bias. Under null, it slightly worsens predictions. Scaling reduces both effects. Only the two common-signal primary intervals clear the prospective 10^{-4}

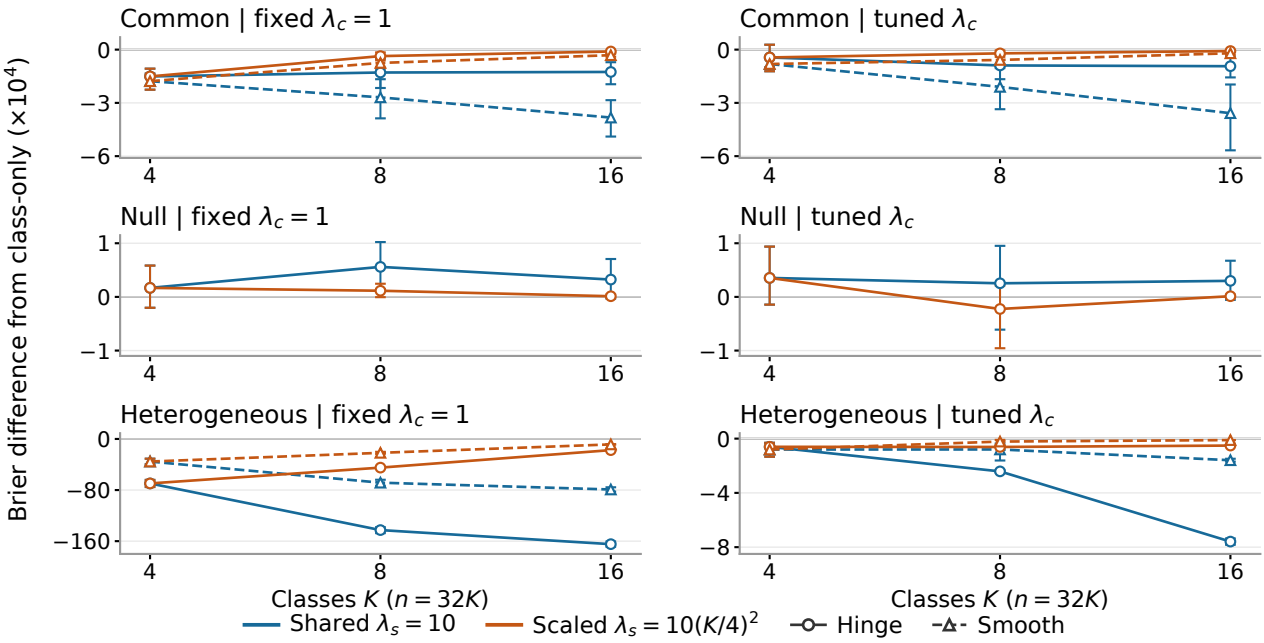


Figure 1: **Effects by class count and truth shape.** Shared policy minus class-only; negative favors sharing. Bars: secondary pointwise 95% intervals over 24 paired seeds, not simultaneous intervals. Null shapes coincide and are drawn once; policies coincide at $K = 4$. Common/null axes match within rows; heterogeneous panels use different ranges. All values have the same 10^4 multiplier. Since $n = 32K$, trends do not isolate class count alone.

reporting scale, which is not a validated utility threshold.

Class count and truth shape matter. Figure 1 separates both shapes and policies. Under common signals, fixed sharing benefits every K/shape cell, but only smooth truth shows a clear increase in benefit magnitude with K; hinge effects are nearly flat. Tuned hinge truth at $K = 4$ includes zero. Null effects are not monotone, and only the unscaled fixed $K = 8$ interval excludes zero. Heterogeneous benefits persist for both shapes after tuning, with much smaller magnitudes. These are descriptive secondary patterns, not tests of trend differences. Since $n = 32K$, across-K changes cannot identify a pure class-count effect; within-problem policy comparisons match both K and n.

Independent tuning changes the comparison. Under common signals, the observed gap falls 29%, from 2.06×10^{-4} to 1.46×10^{-4} (1.71% of excess Brier). Both methods' test errors rise: a smaller gap does not mean better predictors. A scalar ridge multiplier still cannot match common and deviation weights simultaneously.

Under heterogeneous signals, the gap falls 97.5%, from 0.009332 to 0.000230. Class-only excess Brier improves from 0.06058 to 0.01944; shared improves from 0.05124 to 0.01921. Most of the fixed-setting advantage disappears as both improve; the smaller remaining gap is still supported by its secondary interval.

A post-hoc paired tuned-minus-fixed gap analysis gives $+6.00 \times 10^{-5}$ with 95% interval $[1.34, 10.35] \times 10^{-5}$ for common signals, and $+0.009102$ with $[0.008906, 0.009293]$ for heterogeneous signals. These directly test effect changes, separately from the four prespecified tests. For null, both the tuned contrast and its change remain unresolved, establishing neither equivalence nor tuning-induced overfitting.

Why zero mean class amplitudes do not imply harm. Sharing helps despite zero mean generating amplitudes. Balancing d_c cancels coefficients, not probability changes: each multiplies a different class score-gap feature, softmax couples classes, and affine corrections are jointly fitted. Risk need not separate into common and deviation squared errors, so the scalar no-signal conclusion does not follow. This identifies a limit of the surrogate; it does not identify the cause of the benefit. We did not experimentally isolate these interactions.

5 Scope and Numerical Checks

Real-data scope. CalArena [2] has 624 pairs over 78 datasets. Original Brier improved from 0.210593

Table 2: **Which component explains the gain?** Test-Brier differences, scaled by 10^4, with pointwise dataset-block 95% intervals. Negative HSSC–control differences favor HSSC. Each method selects its own candidates.

Question tested	Control	Candidates	Mean difference	95% interval
CalArena: 624 model–dataset pairs, 78 distinct datasets, three equally weighted suites				
Whole-procedure gain	SMS (original)	3	-2.29	$[-4.15, -0.71]$
Sharing at original penalties	Class-only	3	-0.001	$[-0.266, 0.263]$
Whole procedure with wider tuning	SMS	15	-1.42	$[-8.19, 5.33]$
Sharing with wider tuning	Class-only	15	-0.39	$[-1.20, 0.12]$
Projection contribution	No-projection	15	0.0005	$[-0.0049, 0.0056]$
Effect of widening the search	SMS gap change[†]	$3 \to 15$	$+0.87$	$[-6.16, 8.51]$
Development-unseen data: five datasets; 50 dependent model/repeat settings				
External reproduction	SMS	15	-0.006	$[-1.74, 1.27]$

[†] Paired change $(\text{HSSC} - \text{SMS})_{15} - (\text{HSSC} - \text{SMS})_3$; all other rows are HSSC minus control. The original SMS row is frozen; the remaining analyses are post-review. External intervals use five equally weighted dataset means.

to 0.210364 (0.11%). HSSC outperforms SMS on 359 of 624 pairs. Sharing attribution and broader SMS comparisons remain unresolved (Table 2). Knots and penalties used its calibration data; later controls followed outcome inspection.

Three-fold Brier with fold-local TS selects each method's settings. Wider grids cross three affine with five class penalties; SMS gets 15 affine candidates. Intervals use 10,000 paired dataset-block draws and three equally weighted suites. External intervals resample five development-unseen dataset means, not 50 independent settings. HSSC wins 2/5 means and 21/50 settings; calibration evidence is mixed. Fit/inference cost ratios versus SMS are 1.47–11.07/1.60–3.14. General or practical superiority is unestablished.

Numerics as a supporting check. All 19,530 synthetic fits meet the residual-based 10^{-8} Brier-error target (maximum 9.37×10^{-11}); separate coefficient evaluation agrees within 3.34×10^{-16}. Float64 bounds are conditional, not rounding certificates. Refinement on 18 selected problems reduced equivalent-model disagreement from 3.38×10^{-6} to 2.54×10^{-15}. Two rank reversals against class-only end at 3.63×10^{-8} and 2.09×10^{-7}, establishing no consequential selection error or revised 624-pair aggregate.

Submitted supplementary files. `supplement.pdf`: definitions/proofs (§§2–4), numerics (§§5–6), synthetic settings/results (§14), paired tuning (§15). Code/records: `review_artifact.zip`; combined documents: `review_packet_with_lean.pdf`. The 56 Lean lemmas are conditional, excluding experiments, analytic NLL/softmax premises, floating-point execution and statistical intervals.

Conclusion. Sharing weakens common-component regularization without expanding the function family. Tuning narrows our synthetic advantage, while heterogeneous gains remain unexplained and real-data superiority remains unestablished.

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